

BEYOND THE BASE: FREQUENCY ALLOCATION IN ROPE

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ABSTRACT

Rotary position embeddings (RoPE) typically tie their frequency placement to a single base. We study frequency allocation as a design dimension for model quality within and beyond the native context window. Controlled interventions show that internal placement matters even when the frequency range and total log-frequency displacement are fixed. Full sine-cosine geometry characterizes positional overlap, while coordinate interventions distinguish this structure from the model’s learned use of frequencies. We introduce TailSpline by formulating and solving a discrete allocation problem for the transition into the extended low-frequency tail. Its closed-form rule retains standard rotary computation without weight updates or calibration. Matched evaluations show strong long-context task gains across model families, including natural question answering, with a slight native-window reduction. A complementary native-window construction improves in-window task performance using only public RoPE parameters. Cosh transport supplies supporting learning and extrapolation evidence. Together, these results establish internal frequency placement as a practical design dimension for both context extension and in-window quality.

1 INTRODUCTION

For a model trained to length L , sL specifies a target window; quality within that window determines its usefulness. RoPE’s base sets a frequency range, while internal placement determines the positional basis. We study how this allocation affects model quality.

Frequency design includes frozen scaling policies such as YaRN and MrRoPE and learned frequencies such as LeRoPE (Peng et al., 2024; Tian et al., 2026; Karypis et al., 2026). Our paired interventions show that interior placement improves performance at fixed frequency endpoints. Further controls isolate shape effects at equal total displacement and distinguish positional structure from learned coordinate use.

We turn this design freedom into explicit constructions. TailSpline connects the transition to the extended low-frequency tail, improving long-context tasks across model families and on natural question answering. Native Contrastive Proximal allocation (NCP) improves task performance within the original window. Cosh supplies complementary evidence through paired learning and extrapolation.

Our contributions are:

- **Range and allocation have distinct, interacting effects.** At fixed endpoints, internal placement improves performance; changing the range can reverse the preferred allocation.
- **Positional structure and learned use are distinct.** Complete rotary pairs characterize positional overlap; coordinate interventions and rank-quality comparisons distinguish it from learned frequency use.
- **Explicit allocations improve extension and in-window quality.** TailSpline provides a closed-form extension with cross-model and natural-QA gains; NCP improves native-window tasks at unchanged endpoints.

2 ALLOCATION AS A DESIGN VARIABLE

For $K = d_{\text{rot}}/2$ pairs, standard RoPE uses $\omega_k = b^{-k/K}$. Any strictly ordered positive table admits

$$x_k = -\log \omega_k = a + Rz_k, \quad z_k = \frac{x_k - x_0}{x_{K-1} - x_0}, \quad z_0 = 0, \quad z_{K-1} = 1. \quad (1)$$

The endpoints (a, R) fix the sampled log-frequency interval, called the support; $K - 2$ interior coordinates define allocation. Every geometric table has $z_k = k/(K - 1)$ irrespective of base. The full space $\mathcal{Z}_K = \{0 = z_0 < \dots < z_{K-1} = 1\}$ includes analytic subfamilies and positive-gap learned parameterizations.

Allocation applies to a chosen frequency range, whether that range is retained or extended. For deployment relative to a native table, write

$$\omega'_k = \omega_k^N s^{-m_k}, \quad d_k = \log(\omega_k^N / \omega'_k) = m_k \log s. \quad (2)$$

Changing frequencies preserves RoPE’s relative-position composition and norm preservation (Su et al., 2024). Multiplying cosine/sine amplitudes by g separately scales the rotary Q/K contribution by g^2 . Fixed-support experiments isolate z ; complete methods may jointly choose allocation, range and amplitude.

Geo denotes each protocol’s geometric reference and Native a released model’s inherited table. Standard and midpoint geometric grids share z but differ by an overall frequency offset. Exact grid and endpoint conventions are in Appendix B.3.

Separating the effects of allocation. Fixed endpoints isolate internal placement. Holding $D = \sum_k d_k$ fixed then compares allocations with equal total log displacement; permuting a fixed frequency multiset tests coordinate assignment. At fixed native table and target endpoints, D is itself a statistic of z . Equal-displacement controls therefore resolve finer configuration effects within this design space. The complete control definitions are in Appendix C.

3 FREQUENCY ALLOCATION MATTERS AT FIXED RANGE

We first isolate the effect of interior placement by fixing the frequency endpoints. We then examine how that effect depends on training, total frequency displacement, and the model’s learned use of its coordinates.

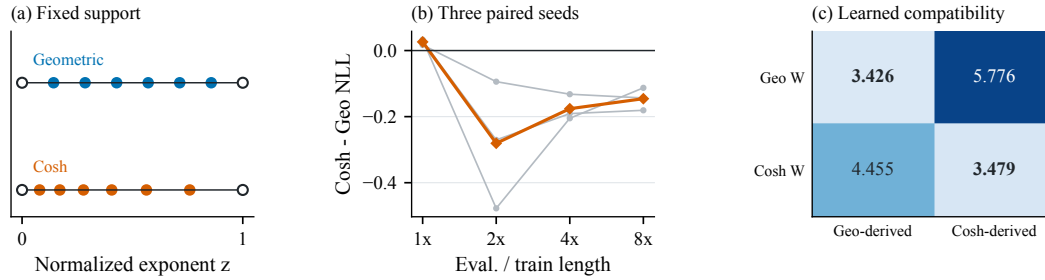


Figure 1: **Fixed-support allocation and learned compatibility.** (a) Fixed endpoints leave internal spacing free ($K = 8$ illustration). (b) Three paired training seeds favor Cosh at extended lengths with retained support; native NLL increases. (c) Two-seed mean tail NLL for the 151.9M weights-by-derived-table crossing at 1K. Rows freeze trained weights; columns install runtime tables. The matched diagonals outperform the crossed installations.

3.1 A PAIRED INTERVENTION AT FIXED SUPPORT

Reallocating interior frequencies improves learned extrapolation at the same frequency endpoints. We train paired 151.9M models with $K = 32$ rotary pairs at length 256.

Within each of three seeds, architecture, initialization, token order, optimizer, schedule, and the 499,974,144-token budget are identical. The matched geometric control (Geo) uses the training-base configuration $b = L_{\text{train}}$ from FMRoPE (Oka et al., 2026); the EVQ-Cosh intervention anchors its quantiles with $\tau = 4$ to the same highest and lowest frequencies and log-span (§5.2). Only the 30 interior z coordinates differ.

With the training support retained at evaluation, mean Cosh-minus-Geo tail NLL is $+0.026$ at the training length and $-0.281 / -0.176 / -0.146$ at 512/1024/2048 tokens. All three paired seeds improve at every extended length (Fig. 1b).

Retargeting both supports to $b = L_{\text{eval}}$ while retaining each trained checkpoint and normalized allocation reverses the ordering: Cosh-minus-Geo NLL becomes $+0.060 / +0.227 / +0.460$, with all three seeds favoring Geo (Table 6). Allocation is a useful design variable, and its preferred setting depends on the support policy.

The effect extends beyond one allocation family. A 50.9M short-training factorial (8.39M tokens per arm) with three paired seeds tests two bases, two training lengths and three head dimensions. Reference Cosh, its preassigned $1.25\times$ strength, and a displacement-matched exponential improve OOD NLL in 7/12, 10/12, and 9/12 configurations. Cosh and the matched exponential perform similarly (mean difference $+0.00074$ NLL, interval $[-0.006, 0.008]$). These results support nonuniform allocation across families, with strength-dependent trade-offs; full settings and a separate standard-grid control are in Appendices C.4 and C.4.

3.2 THE EFFECT REMAINS IN FROZEN MODELS

On OLMo-2-1B and Qwen-2.5-1.5B, we fix the checkpoint, sampled endpoints, 64 pairs, attention amplitude, input rows, and decoder. The range extends from the native fast frequency to one-quarter of the native slow frequency. On OLMo’s nine held-out RULER tasks (Hsieh et al., 2024), uniform normalized exponents score 0.56%, a coarse ramp 61.04%, and a residual-guided profile 60.47%. Qwen’s four-task development comparison gives 57.75/64.00/66.50% (Appendix C.5). Both reallocations yield large OLMo gains while retaining the same sampled range.

Internal shape beyond total displacement. Fixed endpoints do not fix the sum of native-relative shifts $\sum_k d_k$. A separate frozen OLMo comparison holds that sum, endpoints, band and gain fixed: the symmetric profile BM and uniform transition MrRoPE-Uni score 51.32 and 32.12% in a six-task, 48-input 16K follow-up comparison. Total displacement therefore does not explain this score difference (Appendix C.5). The clean TailSpline-C comparison confirms a shape effect at matched displacement on Llama (Section 6.1).

A learned model uses more than a frequency range. Crossing the runtime tables of the paired 151.9M models tests whether weights can use a different allocation interchangeably. Both seeds favor the table derived from their training allocation (Fig. 1c; Appendix C.3). A complementary frozen intervention permutes the same frequency multiset across coordinates and changes model performance (Appendix C.6). These findings connect the two operating settings: learning can adapt weights to an allocation, whereas frozen extension must work with the model’s existing frequency-coordinate association.

4 POSITIONAL STRUCTURE AND CONTENT USE

The controlled gains raise a structural question: what changes when the frequency range stays fixed? We examine the positional directions supplied by complete rotary pairs. Their overlap connects frequency spacing to redundancy within a finite window; the model learns how to combine these directions with content.

4.1 COMPLETE ROTARY PAIRS AND POSITIONAL OVERLAP

For a fixed content pair, one rotary channel contributes $C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to an attention logit, where Δ is relative token separation. Both coefficients depend on content; the positional object is

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}. \quad (3)$$

To avoid privileging a content phase, let $x_\omega(\Delta) = [\cos(\omega\Delta), \sin(\omega\Delta)]$ and define $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$, $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ under a declared separation measure. For nonsingular self-Grams, the singular values of $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ are the canonical correlations, or cosines of principal angles, of the two spaces (Björck & Golub, 1973). Their mean square,

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1], \quad (4)$$

measures shared positional directions and is invariant to invertible basis changes within either pair. Let Γ be the Gram matrix of the concatenated block-whitened pairs, with diagonal blocks I_2 . The average pairwise overlap $\bar{c}$ gives the exact Rényi-2 effective-rank identity

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (5)$$

Here $\text{tr}(\Gamma^2)$ is the frame potential of the normalized positional functions (Benedetto & Fickus, 2003). The identity specializes this classical measure to full rotary pairs: increasing pairwise overlap concentrates the same $2K$ coordinates into fewer effective positional directions. These are normalized directions in the supplied frequency basis; content-dependent coefficients determine their contribution to attention logits.

4.2 FINITE-WINDOW CONCENTRATION OF SLOW DIRECTIONS

Proposition 1 (Shared slow-frequency subspace). *Under uniform measure on $[0, L]$, with $t = \Delta/L$, as $x = \omega L \rightarrow 0$, $\cos(xt) \rightarrow 1$ and $\sin(xt)/x \rightarrow t$, so V_ω converges to $\text{span}\{1, \Delta\}$. For $\epsilon = \max\{|\omega L|, |\nu L|\} \rightarrow 0$, squared chordal separation satisfies $2 - \|Q_{\omega\nu}\|_F^2 = \mathcal{O}(\epsilon^4)$.*

Distinct slow frequencies therefore approach a shared two-dimensional positional space. On the identification grid $b = 256$, $K = 32$, $L = 256$, the eight slowest pairs span $\omega L \in [1.19, 4]$ and have effective rank $r_2 = 2.11$. Their overlap is substantial even beyond the small-phase limit (Fig. 2; Appendix A.2).

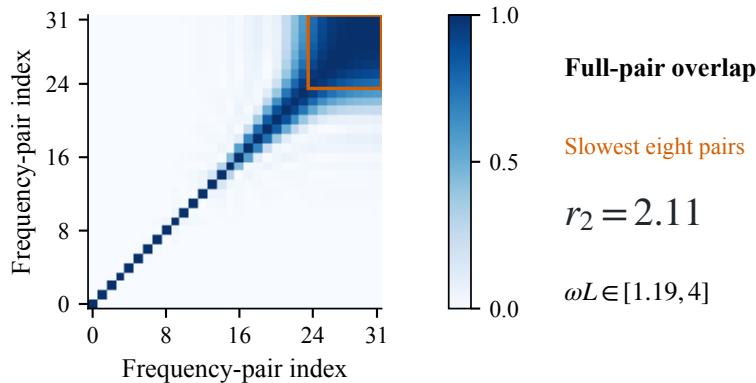


Figure 2: **Allocation determines the supplied positional directions.** Complete-pair overlap on the identification grid. The eight slowest pairs occupy 16 coordinates but have normalized positional effective rank 2.11 under uniform separations; the outlined block marks these pairs.

Why use complete rotary pairs? A cosine-only overlap proxy can reverse the allocation ordering: explicit four-pair tables with identical endpoints give $C_{\cos}(A) = 0 < C_{\cos}(B)$, yet $r_2(A) = 6.29 < 8.00 \approx r_2(B)$ because sine-cosine cross terms also carry positional overlap (Appendix A.3). The complete-pair measure captures this structure independently of phase.

4.3 SHARED POSITIONAL DEPENDENCE AND CONTENT USE

Each rotary block remains invertible even when its positional functions overlap. Shared dependence on distance can therefore coexist with distinct content comparisons. A slow-block construction makes this distinction explicit (Appendix A.5), complementing prior observations of positional and semantic frequency use (Barbero et al., 2025).

The complete frozen Llama tables provide a concrete example. At 16/32K, TailSpline has full-pair effective ranks 8.28/10.08, compared with MrPro’s 8.74/10.21 under uniform separations, while improving clean RULER by 3.39/11.72 points (Table 1). The formulas, full precision and alternative separation measures are in Appendix A.6. These gains establish the value of controlling distance dependence through allocation even when normalized positional diversity decreases. The model’s use of the supplied frequencies determines their practical value.

Frequency changes and coordinate assignment. Changing frequencies also differs from reassigning a fixed spectrum. In the no-alias interval $(0, \pi)$, position-independent invertible Q/K maps preserve the full integer-position rotary kernel exactly if and only if the frequency multisets agree (Appendix A.4). This criterion complements the coordinate interventions and STRING’s structural characterization (Schenck et al., 2025).

5 EXPLICIT FREQUENCY ALLOCATIONS

Allocation changes how a fixed number of rotary pairs covers the frequency range. Our main frozen construction, TailSpline, designs this change around the model’s native table. Cosh supplies a complementary transport for paired training; Section 6.2 develops a native-window allocation.

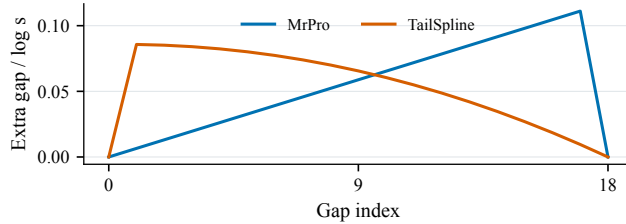


Figure 3: **TailSpline connects the transition to the extended tail.** Extra log-frequency gaps for $n = 17$. Both profiles retain the same outer bands; TailSpline allocates a larger entry gap and smoothly approaches the zero extra gap of the fully interpolated tail.

5.1 TAILSPINE: ALLOCATING EXTENSION AROUND A NATIVE TABLE

A frozen model already uses its native frequencies. We retain its high-frequency band and divide its low-frequency tail by s , while designing the intervening allocation. Following the 32/1-turn boundaries used by MrRoPE (Tian et al., 2026), let l be the last pair making more than 32 turns over the native window and h the first making less than one. For a nonempty transition, put $n = h - l \geq 1$ and $q = \text{clip}(k - l, 0, n)$. The transition accumulates increments $\epsilon_q = m_q - m_{q-1}$, with $m_0 = 0$, $m_n = 1$.

The increments have a direct spectral meaning. For a geometric native grid with gap $c = \log b/K$,

$$x'_{l+q} - x'_{l+q-1} = c + \epsilon_q \log s. \quad (6)$$

Their unit sum allocates exactly $\log s$ of extra span across the band. At pair q , a fixed exponent m_q assigns wavelength growth s^{m_q} ; the endpoints fix the total span, while the interior profile determines how this growth is distributed across channels. The outer bands preserve two distance references: high frequencies retain $R_{\omega^N}(d)$, while the low-frequency tail satisfies $R_{\omega^N/s}(sd) = R_{\omega^N}(d)$. At the tail, constant displacement $m = 1$ preserves native adjacent gaps, so the extra gap returns to zero.

MrRoPE-Pro uses increasing increments, $m_q = q(q+1)/[n(n+1)]$. We formulate a discrete tail-connection problem by penalizing variation in the extra log gaps and their mismatch with the

fully interpolated tail:

$$\min_{\epsilon: \sum_{q=1}^n \epsilon_q = 1} \sum_{q=1}^{n-1} (\epsilon_{q+1} - \epsilon_q)^2 + \epsilon_n^2. \quad (7)$$

The first term smooths neighboring extra gaps; the second closes the transition at the zero-extra-gap boundary supplied by the fully interpolated tail. The high-frequency outer band is fixed by construction. Leaving the entry unpenalized allows a larger first extra gap while prioritizing the smooth tail connection.

Theorem 2 (Finite-grid TailSpline). *The unique minimizer of Eq. (7) has positive increments and cumulative profile*

$$\epsilon_q = \frac{3(n+q)(n-q+1)}{n(n+1)(2n+1)}, \quad m_q^{\text{TS}} = \frac{q(3n^2 + 3n + 1 - q^2)}{n(n+1)(2n+1)}. \quad (8)$$

Strict convexity and the tridiagonal stationarity system give Eq. (8) (Appendix B.1). Adding the entry penalty ϵ_1^2 yields the symmetric construction BM, with increments proportional to $q(n-q+1)$ (Appendix B.1). Relative to MrPro, TailSpline accepts a larger entry jump and reduces the terminal jump by the exact factor $3/(2n+1)$: 0.086 for Llama’s $n = 17$ and 0.081 for OLMo’s $n = 18$ (Fig. 3). TailSpline also redistributes scaling across the entire transition: its wavelength multiplier is $s^{m_q^{\text{TS}}}$, with $m_q^{\text{TS}} \geq m_q^{\text{Pro}}$ at every interior pair. On the Llama $s = 4$ grid, the largest TailSpline/MrPro wavelength ratio is 1.766. Appendix B.4 derives this ordering and compares the response to scale with YaRN’s frequency blend.

What the comparison isolates. TailSpline and MrPro retain the same outer bands and endpoints, so their comparison tests complete internal allocations. At fixed support, total log-frequency displacement is a statistic of z . The control C also matches this statistic to study the remaining shape difference (Appendices B.2 and D.3).

Installation. (1) Read the public native grid, reference length and scale s ; (2) compute the 32/1-turn band; (3) evaluate $m_k, \omega'_k = \omega_k^N s^{-m_k}$ and $g = 1 + 0.1 \ln s$; (4) install before prefill, retaining one table across layers and input lengths. The baseline shares g . Construction costs $O(K)$ and leaves the rotary computation unchanged. Neither constructor reads weights, activations or calibration outputs.

5.2 COSH AS A SUPPORTING EXTRAPOLATION TRANSPORT

Cosh changes z through a monotone map of normalized exponents. Motivated by finite-window overlap among slow pairs, it balances frequency concentration against slow-end tail mass. The density objective and its unique minimizer are derived in Appendix B.3. Its inverse CDF is

$$Q_\tau(u) = 1 - \frac{\text{arcsinh}((1-u) \sinh \tau)}{\tau}, \quad u_k = \frac{k + 1/2}{K}. \quad (9)$$

The uniform-density limit is recovered as $\tau \rightarrow 0$. Affine anchoring of $Q_\tau(u_k)$ to the chosen endpoints gives a convex allocation that moves interior frequencies toward the faster end; Appendix B.3 gives the installed grid and Eq. (31). This is the transport used in the fixed-range training intervention of §3.1. The strength τ determines the amount moved: the fixed-support reference is $d_{\text{head}}/\sqrt{L_{\text{train}}}$, and the MLA extrapolation recipe uses 1.414. The factorial compares neighboring strengths and a matched exponential transport (Appendix C.4). Every installation retains the standard rotary operator.

6 MODEL QUALITY WITHIN AND BEYOND THE NATIVE WINDOW

We evaluate one static TailSpline table across context lengths, test internal shape with an equal-displacement control, and examine in-window allocation with NCP. Supporting Cosh experiments study allocation during learning. The protocol glossary in Table 3 distinguishes these interventions.

6.1 TASK QUALITY WITH A FROZEN TAILSPLINE TABLE

Llama-3-8B-Instruct uses $s = 4$, native length 8192, band $[18, 35]$ and gain $1 + 0.1 \ln 4$. OLMo-2-1B-Instruct uses native length 4096 and band $[14, 32]$ with the same scale and gain rule. T/P denote TailSpline and MrRoPE-Pro. Each pair shares weights, outer frequencies, endpoints, gain, inputs and decoder; only its internal allocation changes. Clean RULER (Hsieh et al., 2024) uses source-order inputs, no content padding or depth selection, batch 1 and the official scorer. Table 1 reports scores and paired 95% intervals.

Table 1: **Quality across context lengths and model families.** Clean Full-13 task-equal scores. One static $s = 4$ table per model; paired intervals use each panel’s inputs. Original RoPE scores 90.03% on the Llama 8K reference.

Model	Length	N/task	T (%)	P (%)	Δ (pp)	95% interval
Llama	8K (L)	50	85.16	82.39	+2.76	[1.29, 4.25]
	16K ($2L$)	50	86.09	82.71	+3.39	[1.53, 5.34]
	32K ($4L$)	200	68.27	56.54	+11.72	[10.32, 13.11]
OLMo	16K ($4L$)	200	50.65	9.23	+41.42	[39.99, 42.81]

Quality across the window. The Llama 32K panel contains 2,600 paired prompts, with actual lengths 28,270–32,606 tokens. Its +11.72-point gain spans all four task families; 12 of 13 task means improve. The 16K panel adds 650 pairs and a +3.39-point gain. RULER QA improves by 8 and 9 points at 16K and 32K, respectively. The same table serves native-length inputs: on 650 clean 8K prompts, it scores 85.16%, with a slight reduction from original RoPE’s 90.03%. Full native and extended task breakdowns are in Appendices D.2, D.2 and D.2.

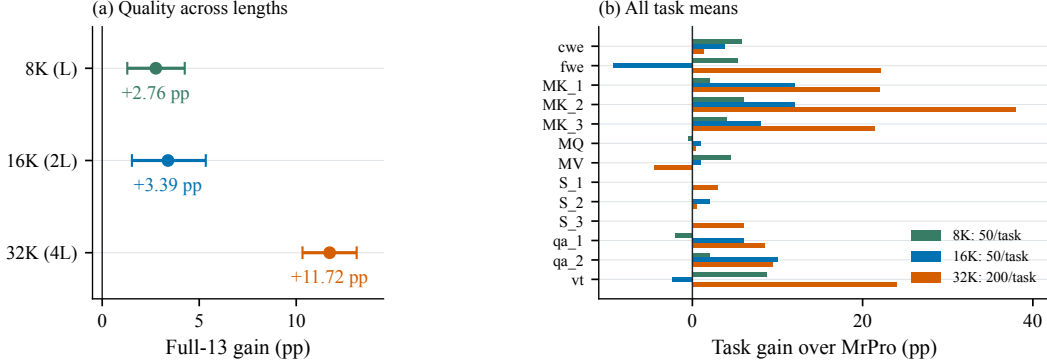


Figure 4: **One static allocation improves quality across context lengths.** Clean Llama Full-13 T-P differences. (a) Task-equal scores with paired 95% intervals at L , $2L$ and $4L$. (b) All task differences at each length. The 8/16K panels use 50 inputs per task and 32K uses 200.

Cross-model confirmation. OLMo’s clean 16K panel contains 2,600 pairs and improves by 41.42 points, with positive differences on all 13 tasks. Retrieval, tracking, aggregation and QA improve by 60.19, 9.70, 16.13 and 7.50 points, respectively; each family interval is positive. This confirms the allocation advantage on a second model family at its target window (Appendix D.2). Classic multi-length task and PPL curves remain in Appendix B.1; these retain their original sampling protocols.

Natural question answering. The frozen five-task Natural-QA pool contains 631 questions from 524 source-context clusters. Each model retokenizes the same source pool; OLMo’s actual inputs are 4,110–16,319 tokens, all beyond its native window. Task-macro whole-response F1 improves by 3.30 points, with positive mean differences on all five tasks. Llama’s F1 scores are close (Table 2). LongBench v2 (Bai et al., 2025) extends the Llama evaluation to 117 complete inputs between 8K

and 32K, using its multiple-choice accuracy metric. Source-context clustered intervals, per-task results and output statistics are in Appendices D.4 and D.4.

Table 2: **Natural-task evaluation.** Natural-QA reports five-task macro F1; LongBench v2 reports accuracy on complete inputs in the stated token range. Scores are percentages, with source-context clustered intervals.

Model	Benchmark	N	T	P	Δ (pp)	95% interval
Llama	Natural-QA	631	41.08	40.88	+0.20	$[-1.53, 1.89]$
OLMo	Natural-QA	631	24.92	21.62	+3.30	$[0.34, 6.40]$
Llama	LongBench v2	117	35.04	30.77	+4.27	$[-3.39, 11.97]$

Internal shape at equal displacement. The clean T-C comparison matches frequency endpoints, outer bands, gain and total log-frequency displacement. At 32K, T/C score 68.27/66.17%: a +2.10-point difference with interval $[1.11, 3.08]$. At 16K, they score 86.09/86.35%. Thus, total displacement does not fully determine allocation quality; the internal shape matters at the target length (Appendix D.3). C has slightly higher positional effective rank than T at 32K (10.093/10.077), complementing the distinction between positional structure and learned use in Section 4.

6.2 IMPROVING TASK PERFORMANCE WITHIN ROPE’S NATIVE CONTEXT

Optimizing z improves task performance within RoPE’s native context. We construct *Native Contrastive Proximal* allocation (NCP) from the public native grid and length. A binary reference comparison models an aligned target competing with a uniformly phased distractor. Averaging its logistic loss over relative separations gives a reference risk $\mathcal{R}(\phi)$ at phase span $\phi = (L - 1)\omega$. NCP selects log-frequency shifts u_k through

$$\min_u \sum_k \left[\mathcal{R}(\phi_k e^{-u_k}) + \frac{9}{8} u_k^2 \right], \quad \omega'_k = \omega_k e^{-u_k}. \quad (10)$$

The quadratic term keeps the allocation near its native grid. We fix both endpoints, bound shifts, and preserve frequency ordering and minimum log gaps; Appendix E.1 gives the reference model, constraints and numerical construction. Gain remains 1, and the same table is installed across layers and inputs without weight updates or calibration.

On OLMo’s native 4K clean Full-13 panel (780 paired prompts), NCP raises the score from 69.37 to 70.78%, a +1.41-point gain with paired interval $[0.20, 2.58]$. This gain establishes a direct in-window benefit from optimizing frequency allocation at fixed support. Appendix E.2 reports the complete task breakdown and companion direction controls.

6.3 SUPPORTING EXTRAPOLATION EVIDENCE WITH COSH

A 432M MLA-style model uses 16 rotary pairs, 500M training tokens at 8K and three seeds (DeepSeek-AI, 2024a;b), with fixed Cosh strength $\tau = 1.414$. All arms use the same FineWeb-Edu evaluation chunks. This recipe uses unanchored midpoint tables, distinct from the fixed-endpoint intervention. Cosh improves all five extended lengths in every seed; 16K PPL falls from 138.8 to 95.6. The benefit appears during training and persists under a common wavelength blend (Fig. 5; Appendix F.1). A 750M single-seed continuation fixes endpoints and reduces 16K PPL from 45.1 to 24.4; 8K answer-token exact match rises from 0/40 to 31/40. Full recipes and matched OLMo adaptation are in Appendix F.2.

7 RELATED WORK

Frequency scaling and frozen deployment. RoPE (Su et al., 2024), base analyses and FMRoPE connect frequency range with extrapolation (Liu et al., 2024; Xu et al., 2024; Oka et al., 2026). Position interpolation, NTK scaling, YaRN and MrRoPE provide established rescaling policies (Chen et al., 2023; bloc97, 2023; Peng et al., 2024; Tian et al., 2026). MrRoPE’s mixed-radix products

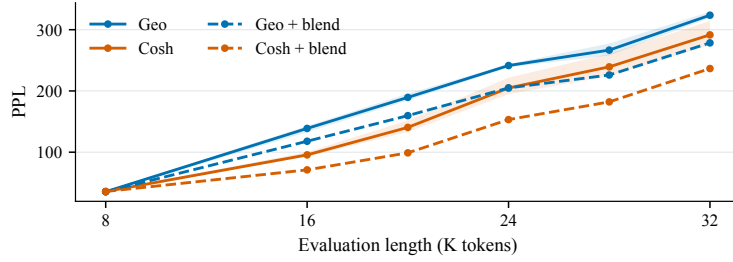


Figure 5: **Cosh improves extrapolation across training seeds.** Three-seed mean and range for the 432M MLA model; dashed curves use the common wavelength blend ($s = 4$). Evaluation uses common text chunks.

and cumulative log displacement are equivalent coordinates for the same static frequency transform. TailSpline retains its outer bands and changes the transition objective. Our controlled interventions identify the value of the internal allocation, and our discrete objective yields an explicit tail-connection rule. LongRoPE and LongRoPE2 combine search and learning, the latter using mixed-context training (Ding et al., 2024; Shang et al., 2025). Their deployment results have different training budgets from a frozen table swap.

Frequency function and learning. Round and Round studies positional and semantic frequency use; its p -RoPE gains include Gemma training from scratch (Barbero et al., 2025). HoPE studies high-frequency positional encoding (Chen et al., 2025); Massive Values studies large Q/K components (Jin et al., 2025). FoPE changes Fourier components, while LeRoPE learns frequencies and derives their attention/value gradients (Hua et al., 2025; Karypis et al., 2026). We build on frequency design by isolating the value of interior placement and connecting positional structure to explicit constructions for learning, frozen extension and native-window tasks.

Group structure and changes of basis. STRING characterizes differentiable matrix encodings satisfying an identity and a group-like relative-translation law, and relates them to RoPE through an orthogonal basis (Schenck et al., 2025). GRAPE connects group generators, learned frequencies and frequency-norm coupling (Zhang et al., 2026). Our integer-position criterion uses rotation spectra to separate a frequency change from a fixed change of basis (Appendix A.4).

Task structure and positional dependence. PINE changes inter-document masks and ordering to target document-level invariance (Wang et al., 2024); PRoPE encodes relative camera geometry (Li et al., 2025). Selective RoPE and RePo learn input-dependent rotations or positions (Movahedi et al., 2026; Li et al., 2026). These modify different objects from a static frequency allocation. HELMET separates application categories whose behavior synthetic retrieval alone does not predict (Yen et al., 2024), motivating distinct retrieval, natural-QA and native-window evidence rather than a single pooled score.

Our contribution connects controlled allocation interventions, complete-pair positional analysis and explicit constructions. Fixed-range and equal-displacement comparisons establish the design freedom; TailSpline and NCP demonstrate its value for long-context and native-window tasks, respectively.

8 DISCUSSION AND CONCLUSION

A frequency range leaves a consequential choice open: where to place the frequencies within it. Controlled interventions and equal-displacement comparisons establish the value of this choice. Full-pair geometry characterizes positional overlap, while coordinate interventions distinguish it from learned frequency use.

TailSpline improves long-context task quality across model families and on natural question answering. NCP improves native-window tasks at fixed support; Cosh supplies paired learning and extrapolation evidence. Together, these constructions establish z as a practical design dimension for extension and in-window quality.

ETHICS STATEMENT

This work studies frequency allocation in existing language models using public research corpora and evaluation suites under their stated licenses. It introduces no human-subject study or deployed decision system. The proposed constructions retain the rotary operator and add no trainable parameters; adaptation experiments use the weight updates specified in their protocols.

REPRODUCIBILITY STATEMENT

The appendices specify the installed frequency tables, exponent grids, attention amplitudes, training and adaptation recipes, datasets, and evaluation procedures. The fixed-range experiment reports three paired training seeds; the larger-model and frozen-model tables identify their comparison units and sample counts. The source archive provides the active TeX, vector figures, numerical summaries and CPU checks of the Gram-matrix and allocation identities. It also includes the frozen two-arm data-preparation, table-construction, generation, scoring and paired-report entryptoints with their local dependencies. The runtime guide specifies commands, public data sources and the pinned RULER revision.

The core fixed-support identification uses six runs of 499,974,144 tokens each; the 432M comparison uses six 500M-token runs, and the 750M continuation uses two 500M-token runs. These are protocol-specific training budgets, with the corresponding recipes and additional studies detailed in the appendices.

AI USE STATEMENT

We used generative AI tools to assist with mathematical discussion, code development, literature organization, and manuscript editing. Experimental results are drawn from recorded model runs and numerical analyses. The authors are responsible for the correctness and originality of the submitted text, mathematical arguments, experimental claims, and accompanying code.

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SUPPLEMENTARY MATERIAL

The supplement follows the main argument. Appendix A proves the positional statements; Appendix B derives the allocations; Appendix C specifies the controlled interventions. Complete frozen, native-window and learned results are in Appendices D, E and F, respectively. The source archive includes numerical inputs, construction/scoring code and CPU verification scripts.

Table 3: **Protocol glossary.** Each comparison retains its own data, weights, support convention and metric.

Comparison	Matched quantities	Question
Fixed-support training	Initialization, tokens, endpoints	Interior allocation during learning
Frozen T/P	Weights, outer bands, gain, inputs	Complete interior allocations
Frozen T/C	Above plus total displacement	Residual shape
Slot permutation	Weights and frequency multiset	Learned coordinate assignment
NCP/native	Weights, endpoints, unit gain	In-window task improvement
Cosh learning	Recipe, data; grid stated per study	Supporting learned extrapolation

A POSITIONAL GEOMETRY AND LEARNED USE

A.1 CROSS-GRAM FORMULA, INVARIANCE AND EFFECTIVE RANK

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_\star(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (11)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. Thus $c_{\omega\nu}$ is an intrinsic measure of overlap between the two positional subspaces.

Theorem 3 (Spectral budget identity). *For the block-whitened Gram matrix Γ with diagonal blocks I_2 , $\text{tr } \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K-1)\bar{c}]$. Hence its Rényi-2 effective rank is given by (5).*

Proof of Theorem 3. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr } \Gamma = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (5). $\square$

The identity determines Rényi-2 effective rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 1. Put $t = \Delta/L$, $x = \omega L$, and use $\theta = x^2$ as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (12)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (13)$$

Thus $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$. The fourth-order coefficient follows by retaining the part of the first perturbation transverse to V_0 . Under the uniform measure on $t \in [0, 1]$,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (14)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (15)$$

where $\Pi_0^\perp$ is the $L_2[0, 1]$ projection off $\text{span}\{1, t\}$. In the base $[1, t]$, its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (16)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1}M) + O(\epsilon^6) = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6), \quad (17)$$

with $\epsilon = \max\{|x|, |y|\} \rightarrow 0$. The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at $x=0.05, y=0.10$ its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so the constant direction is annihilated before whitening. Writing $\bar{h} = h - \mathbb{E}_p h$, the same expansion gives

$$\frac{\overline{\sin(\omega\Delta)}}{\omega} \longrightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{2\overline{\cos(\omega\Delta)}}{\omega^2} \longrightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (18)$$

These functions span the centred limit whenever p has nondegenerate support on at least three distances. The separation weights p therefore determine which polynomial directions survive centering. $\square$

A.2 FINITE-WINDOW OVERLAP IN THE IDENTIFICATION EXPERIMENT

The slow-limit proposition has a finite-window counterpart that can be calculated directly on the identification grid. Set $b = 256$, $K = 32$, $L = 256$, and retain the n slowest frequencies of $\omega_j = b^{-j/K}$. Their dimensionless frequencies are $\omega_j L = 1.1892, 1.4142, 1.6818, 2, \dots$ in slow-to-fast order. The smallest is already greater than one, so we evaluate the full cross-Gram in (11) without a small-phase approximation.

Table 4: **Finite-window positional concentration on the paired-training grid.** The separation measure is uniform on $[0, L]$. The nominal coordinate count is $2n$; r_2 is the effective rank of the block-whitened Gram.

Slowest pairs n	Largest ωL	Coordinates	r_2
2	1.41	4	2.00030
4	2.00	8	2.00361
6	2.83	12	2.02263
8	4.00	16	2.11474
12	8.00	24	2.89256
16	16.00	32	4.43339

For $n = 8$, the largest two eigenvalues are 7.9681 and 7.5752, accounting for 97.1% of the trace. This is the finite-window geometry in Fig. 2; the full-pair calculation uses no small-phase approximation.

A.3 TWO STATIC ORDERING COUNTEREXAMPLES

Two explicit examples show how phase coupling and interval length change the ordering of frequency tables.

Cosine-only collision can choose the lower-rank table. Take $K = 4$, $L = 8\pi$, and the two descending frequency tables

$$\Omega_A = (1, 5/8, 3/8, 1/4), \quad \Omega_B = (1, 6.01/8, 4.01/8, 1/4).$$

They share both endpoints. Define $C_{\cos}$ as the mean squared normalized cosine-cosine overlap over distinct pairs. For Ω_A , every cosine is an integer harmonic on $[0, L]$, so $C_{\cos}(A) = 0$ exactly. Mixed harmonic parities nevertheless produce nonzero sine-cosine cross terms. For Ω_B , the small perturbation of an even-harmonic table gives

$$C_{\cos}(A) = 0 < 1.38725 \times 10^{-5} = C_{\cos}(B), \quad (19)$$

$$r_2(A) = 6.29478 < 7.99971 = r_2(B). \quad (20)$$

Thus a strict cosine-only preference can have the opposite full-subspace rank ordering. Both values follow directly from Eq. (11).

Full-subspace collision can reverse across lengths. Now use $K = 4$, $L = 2\pi$, and the fixed tables

$$\Omega_A = (1, 0.99, 0.02, 0.01), \quad \Omega_B = (1, 2/3, 1/3, 0.01).$$

Again the endpoints agree. Writing C_M for mean phase-invariant collision under the uniform measure on $[0, M]$, Eq. (11) gives

$$C_L(A) = 0.538996 < 0.631721 = C_L(B), \quad (21)$$

$$C_{2L}(A) = 0.384445 > 0.204995 = C_{2L}(B), \quad (22)$$

$$C_{4L}(A) = 0.345188 > 0.033506 = C_{4L}(B). \quad (23)$$

The close pairs in A retain high overlap as the interval grows, while the more evenly spread frequencies in B become easier to distinguish. The example makes the interval dependence of the geometry explicit.

All frequency entries and measures are specified above. The accompanying `figs/verify_explicit_geometry.py` evaluates the closed-form Gram and checks it against independent 128-point Gauss-Legendre quadrature; the largest discrepancy in the displayed metrics is below 4×10^{-14} . These finite basis examples complement the trained allocation comparisons and the weight-table crossings.

A.4 INTEGER-POSITION KERNEL EQUIVALENCE AND ITS BOUNDARIES

Corollary 4 (Integer-position equivalence). *Let Ω, Λ contain K frequencies in $(0, \pi)$ and $\mathcal{R}_\Omega(d) = \text{diag}_k R(\omega_k d)$. Position-independent invertible A, B satisfy $A^\top \mathcal{R}_\Omega(d) B = \mathcal{R}_\Lambda(d)$ for $d = 0, 1$ if and only if the frequency multisets agree.*

Corollary 4 applies to the full bilinear kernel on the $2K$ -dimensional rotary space. Equality for all q, k is matrix equality. At $d = 0$, $A^\top B = I$; at $d = 1$ the two rotation matrices are similar. Their complex eigenvalues are $e^{i\omega_k}, e^{-i\omega_k}$ including multiplicities. On $(0, \pi)$ the positive-angle eigenvalue identifies each frequency uniquely, proving necessity even with repeated frequencies. For sufficiency choose a block permutation P with $\mathcal{R}_\Lambda(d) = P \mathcal{R}_\Omega(d) P^\top$, and set $A = B = P^\top$. This holds for every integer d , including negative distances.

The interval $(0, \pi)$ selects unique positive rotation angles: outside it, ω and $\omega + 2\pi$ are indistinguishable at integer positions. The statement concerns equality of the full bilinear kernel for every content pair. These conditions are verified by the bundled discrete-kernel checker.

A.5 POSITIONAL OVERLAP PRESERVES CONTENT COORDINATES

For a block $\mathcal{B}$ of r rotary pairs, write $\mathcal{R}_\mathcal{B}(d) = \text{diag}_{k \in \mathcal{B}} R(\omega_k d)$. Each block is orthogonal, so $\mathcal{R}_\mathcal{B}(d)$ has rank $2r$ at every distance. This is the rank of an operator on content coordinates; the positional Gram instead measures shared dependence on distance.

Proposition 5 (Content matching in a slow block). *Let $H > 0$ and $\eta = H \max_{k \in \mathcal{B}} |\omega_k|$. For $|d| \leq H$,*

$$|q_{\mathcal{B}}^{\top} \mathcal{R}_{\mathcal{B}}(d) k_{\mathcal{B}} - q_{\mathcal{B}}^{\top} k_{\mathcal{B}}| \leq \eta \|q_{\mathcal{B}}\|_2 \|k_{\mathcal{B}}\|_2. \quad (24)$$

For $2r$ content symbols represented by distinct coordinate unit vectors, let the query and correct key share a vector and all incorrect keys use other vectors. Their unscaled score margin is at least $1 - 2\eta$ across arbitrary key distances in $[-H, H]$.

Proof. For each pair, $\|R(\omega d) - I\|_2 = 2|\sin(\omega d/2)| \leq |\omega d|$. The block-diagonal norm is the maximum pair norm. The bilinear norm bound gives Eq. (24). Unrotated matching scores are one for the correct symbol and zero for an incorrect symbol. Each score changes by at most η , giving the margin bound. In particular, $\eta < 1/2$ preserves the correct ranking. $\square$

As distinct frequencies approach zero, their positional subspaces approach $\text{span}\{1, d\}$ while the content construction remains distinct in all $2r$ coordinates. This provides a concrete distinction between repeated positional dependence and repeated content information, consistent with prior analyses of frequency use (Barbero et al., 2025). The construction concerns specified content vectors and the small-phase regime; the next subsection evaluates the complete finite frequency tables.

A.6 TASK GAINS WITH LOWER FULL-PAIR EFFECTIVE RANK

We use the Llama mathematical native grid $\omega_k^N = 500000^{-k/64}$, native length 8192, $s = 4$ and band $[18, 35]$. The TailSpline and MrPro formulas determine all frequencies without model observations. Effective rank uses the full-pair Gram from Eq. (11), with uniform separations on $[0, H]$.

Table 5: **A successful allocation can have lower positional effective rank.** Ranks describe the full mathematical frequency tables; task scores reuse the clean paired evaluations in Table 1. The 16K and 32K task panels contain 650 and 2600 paired prompts.

Window	r_2 TailSpline	r_2 MrPro	RULER TailSpline (%)	RULER MrPro (%)
16K	8.283391	8.739887	86.0949	82.7051
32K	10.077270	10.214888	68.2660	56.5436

Continuous and integer-position Grams, FP32 rounding and causal-separation weights retain this rank ordering at 8/16/32K. The table separates positional structure from task quality; a lower rank is not used as a selection rule.

B EXPLICIT CONSTRUCTIONS AND DISTANCE SCALING

B.1 TAILSPLINE AND THE SYMMETRIC CONSTRUCTION

Let $D\epsilon = (\epsilon_{q+1} - \epsilon_q)_{q=1}^{n-1}$ and $H = D^{\top} D + e_n e_n^{\top}$. A zero value of $\epsilon^{\top} H \epsilon$ requires constant increments and a zero terminal increment, hence $\epsilon = 0$. Thus H is positive definite. For $n > 1$, the proposed $\epsilon_q = c(n+q)(n-q+1)$, with $c = 3/[n(n+1)(2n+1)]$, satisfies

$$\epsilon_1 - \epsilon_2 = 2c, \quad 2\epsilon_q - \epsilon_{q-1} - \epsilon_{q+1} = 2c, \quad 2\epsilon_n - \epsilon_{n-1} = 2c.$$

Therefore $H\epsilon = 2c\mathbf{1}$, the equality-constrained stationarity equation. The increments are positive and sum to one; summing them yields Eq. (8). The objective minimum is $2c = 6/[n(n+1)(2n+1)]$. For $n = 1$, the constraint gives $\epsilon_1 = 1$, agreeing with the formulas. Positivity makes additional nonnegativity constraints inactive.

Adding ϵ_1^2 yields the symmetric objective, with unique solution

$$\epsilon_q^{\text{BM}} = \frac{6q(n-q+1)}{n(n+1)(n+2)}, \quad m_q^{\text{BM}} = \frac{q(q+1)(3n+2-2q)}{n(n+1)(n+2)}. \quad (25)$$

Both profiles are monotone with $m_0 = 0, m_n = 1$. The portable CPU check independently solves the constrained tridiagonal system and verifies each closed form.

What the objective smooths. On the geometric native grid, Eq. (6) makes each change between neighboring log gaps equal to $(\epsilon_{q+1} - \epsilon_q) \log s$. The terminal change is $-\epsilon_n \log s$ because the fully interpolated tail has constant displacement. Thus $(\log s)^2$ times Eq. (7) is exactly the squared log-gap variation within the transition and at its low-frequency junction. The high-frequency junction is deliberately unpenalized.

At the terminal gap,

$$\epsilon_n^{\text{Pro}} = \frac{2}{n+1}, \quad \epsilon_n^{\text{TS}} = \frac{6}{(n+1)(2n+1)}, \quad \frac{\epsilon_n^{\text{TS}}}{\epsilon_n^{\text{Pro}}} = \frac{3}{2n+1}.$$

The opposite entry behavior is also explicit: $\epsilon_1^{\text{Pro}} = 2/[n(n+1)]$, whereas $\epsilon_1^{\text{TS}} = 3/(2n+1)$. TailSpline trades a larger entry jump for a smaller terminal jump. A useful boundary prior need not minimize both. In the continuum limit, $m^{\text{TS}}(u) = (3u - u^3)/2$ solves $\min \int_0^1 (m''(u))^2 du$ under $m(0) = 0, m(1) = 1, m'(1) = 0$; the natural entry condition is $m''(0) = 0$. The finite-grid formula is used in every model experiment, not this limiting polynomial.

B.2 AN EXACT EQUAL-DISPLACEMENT CONTROL

Use local indices $q = 0, \dots, n$. Write $T_q = m_q^{\text{TS}}, B_q = m_q^{\text{BM}}, U_q = q/n$, and $P_q = q(q+1)/[n(n+1)]$. Define the front-loaded profile $F_q = 2U_q - P_q$ and the control

$$C_q = (1 - w_n)U_q + w_n F_q, \quad w_n = \frac{3n}{2(2n+1)}. \quad (26)$$

Substitution of the finite-grid formulas gives $T = (1 - w_n)B + w_n F$. Subtracting C gives

$$m^{\text{TS}} - m^C = (1 - w_n)(m^{\text{BM}} - m^{\text{Uni}}). \quad (27)$$

In local coordinates, the explicit difference is

$$T_q - C_q = (1 - w_n)(B_q - U_q) = \frac{q(n-q)(2q-n)}{2n(n+1)(2n+1)}. \quad (28)$$

This cumulative-profile difference is antisymmetric under $q \mapsto n - q$, vanishes at both endpoints, and sums to zero. Since U and F have positive increments and $0 < w_n < 1$, C is monotone with $C_0 = 0, C_n = 1$. Extending both T and C by the same constant outer bands gives identical endpoints and total log displacement at a common scale. Their equal increment centroids follow from Eq. (38), rather than imposing a further constraint. For $n = 1, 2$, $B = U$ and $T = C$; for wider bands the contrast tests higher-order shape.

The clean T-C results are in Appendix D.3. The finite-sum identities are checked with exact rational arithmetic in the bundled allocation constructor.

B.3 COSH DENSITY AND FINITE INSTALLATION

Let $S_\rho(t) = \int_t^1 \rho(\phi) d\phi$. For $\alpha > 0, \beta \geq 0$, the Cosh transport minimizes

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho^2 + \frac{\beta}{2} \int_0^1 S_\rho^2, \quad \rho \geq 0, \quad \int_0^1 \rho = 1. \quad (29)$$

Coercivity and weak lower semicontinuity on the closed convex constraint set give existence; the first term makes the objective strictly convex. For $g(\phi) = \int_0^1 \min(\phi, \psi) \rho(\psi) d\psi$, stationarity is $\alpha\rho + \beta g + \lambda = 0$. Since $g'' = -\rho, g'(1) = 0$ and $g'(0) = 1$, this gives $\rho'' = \tau^2 \rho, \rho'(0) = -\tau^2$ and $\rho'(1) = 0$, where $\tau^2 = \beta/\alpha$. Thus the unique solution is

$$\rho_\tau(\phi) = \frac{\tau \cosh[\tau(1-\phi)]}{\sinh \tau}. \quad (30)$$

It is positive and has unit mass, so it satisfies the inequality constraint. At $\beta = 0$ the solution is uniform. Integrating and inverting its CDF yields Eq. (9). Equivalently, for $h = Q', \mathcal{C}_{\text{app}} = \frac{1}{2} \int_0^1 [\alpha/h + \beta(1-u)^2 h] du$.

For midpoint quantiles $u_k = (k + 1/2)/K$, anchored installation uses

$$z_k = \frac{Q_\tau(u_k) - Q_\tau(u_0)}{Q_\tau(u_{K-1}) - Q_\tau(u_0)} \leq \frac{k}{K-1}, \quad \omega_k = \exp(-a - Rz_k). \quad (31)$$

The inequality follows from convexity of Q_τ and its endpoint chord. Unanchored midpoint installation instead uses $\omega_k = b^{-Q_\tau(u_k)}$. Its uniform limit is midpoint Geo; standard Geo uses k/K , differing by the common frequency factor $b^{-1/(2K)}$. Each experimental protocol below specifies its actual grid and support convention. Native denotes the inherited frequency table of a released model.

B.4 SCALE RESPONSE AND FINITE PHASE CHANGES

Write $\nu_k(s) = \omega_k^N s^{-m_k(s)}$ and let $A_k(s) = \omega_k^N / \nu_k(s)$ be the wavelength multiplier. For the index ramp in the released YaRN implementation (Peng et al., 2024), with fixed interior coordinate $0 < t < 1$,

$$A_Y(s, t) = \frac{1}{1 - t + t/s}, \quad \lim_{s \rightarrow \infty} A_Y(s, t) = \frac{1}{1 - t}. \quad (32)$$

A fixed frequency blend $\gamma + (1 - \gamma)/s$ similarly has limit $1/\gamma$ for $\gamma > 0$; the fully interpolated tail $\gamma = 0$ grows with s . MrPro and TailSpline instead have $A_P = s^{P_q}$ and $A_T = s^{T_q}$ for their fixed finite-grid exponents. Their log-scale derivatives are

$$\frac{\partial \log A_Y}{\partial \log s} = \frac{t}{s(1 - t) + t}, \quad \frac{\partial \log A_P}{\partial \log s} = P_q, \quad \frac{\partial \log A_T}{\partial \log s} = T_q. \quad (33)$$

These identities follow by differentiation. They describe constructor responses to s , rather than task-performance guarantees.

Index and turn-count ramps use different blend weights. All numerical comparisons here use one shared band and the index-ramp equation of the released YaRN implementation (revision 995db5b). They are frequency-table comparisons, not a re-evaluation of the YaRN baseline runs in the MrRoPE paper.

Magnitude of the TailSpline intervention. Direct subtraction of the finite-grid formulas gives

$$T_q - P_q = \frac{q(n - q)(3n + q + 1)}{n(n + 1)(2n + 1)} \geq 0, \quad \frac{A_T}{A_P} = s^{T_q - P_q}. \quad (34)$$

For the Llama $n = 17$ grid, the maximum wavelength ratio is 1.328763, 1.765612, 2.346080 and 3.117385 at $s = 2, 4, 8, 16$. At pair $k = 26$ ($q = 8$), the multipliers are:

Scale	YaRN index blend	MrPro	TailSpline
4	1.545455	1.385674	2.423868
16	1.789474	1.920093	5.875134

The YaRN/MrPro ordering at an interior pair can therefore change with scale. TailSpline redistributes substantial wavelength growth throughout the transition, in addition to its smaller terminal extra-gap jump.

B.5 TWO REFERENCE RESPONSES AND FINITE PHASE INTERVALS

The native reference is ω and the fully interpolated reference is ω/s . For a declared phase tolerance $0 < \theta \leq \pi$, define

$$D_N(\nu; \theta) = \frac{\theta}{|\omega - \nu|}, \quad D_D(\nu; \theta) = \frac{\theta}{|\nu - \omega/s|}, \quad (35)$$

with infinity at a zero denominator. These are the maximal intervals starting at zero on which the respective unwrapped phase deviation is at most θ . Since $\omega/s \leq \nu_T \leq \nu_P \leq \omega$,

$$D_D(\nu_T; \theta) \geq D_D(\nu_P; \theta), \quad D_N(\nu_T; \theta) \leq D_N(\nu_P; \theta). \quad (36)$$

The exact operator difference is $2|\sin(d(\nu - \nu_{\text{ref}})/2)|$. Thus the distances quantify two specified phase references; they are not effective context lengths of a model.

For a fixed content component $f_\nu(d) = a \cos(\nu d - \omega d_0)$, with $a > 0$ and desired stretched distance $d = s d_0$, its residual phase is $\delta = d(\nu - \omega/s)$. If $0 \leq \delta_T \leq \delta_P \leq \pi$, then $f_T(d) \geq f_P(d)$ by the monotonicity of cosine on $[0, \pi]$. The content phase specifies the reference calculation. Outside that interval, reducing residual phase from 2π to π instead changes the response from a to $-a$. The conditional example explains a possible use of distance scaling, while the full-model comparisons measure its actual task value.

Equal displacement does not imply a small phase intervention. On the same Llama $s = 4$ mathematical grid, the largest T/C unwrapped phase difference over integer distances $0, \dots, 32767$ is 10.672514 radians. The corresponding T/P value is 108.197815. For finite comparisons one can use

$$\rho_H(\delta) = \frac{4}{H} \sum_{d=0}^{H-1} \sin^2(d\delta/2) = 2 - 2 \frac{\sin(H\delta/2)}{H \sin(\delta/2)} \cos((H-1)\delta/2), \quad (37)$$

with removable singularities evaluated by continuity. The fixed-slot rotation RMS is $[K^{-1} \sum_k \rho_H(\nu_k - \tilde{\nu}_k)]^{1/2}$. At $H = 32768$, it is 0.448329 for T/C and 0.687543 for T/P. These values compare unscaled rotary kernels, before the shared cosine/sine gain and the attention normalization. For $H|\delta| \ll 1$, $\rho_H(\delta) = ((H-1)(2H-1)/6)\delta^2 + O(H^4\delta^4)$. The small parameter is therefore distance times frequency difference; matching total displacement alone does not supply it.

C CONTROLLED ALLOCATION EXPERIMENTS

C.1 INTERVENTION IDENTITIES

With the native fast endpoint fixed and its slow endpoint divided by s , $z'_k = (Rz_k + \log s m_k)/(R + \log s)$. Fixing the cumulative increment mass $\sum_q \epsilon_q = 1$ fixes the added log span; fixing total displacement $D = \log s \sum_k m_k$ further constrains the interior allocation. In particular,

$$\sum_{q=1}^n q \epsilon_q = n - \sum_{q=1}^{n-1} m_q, \quad D = K a' + R' \sum_k z'_k - \sum_k x_k^N. \quad (38)$$

Thus equal total displacement already fixes the increment centroid. A frequency-multiset permutation instead keeps the spectrum and changes its coordinate assignment. Gain is controlled separately in every matched comparison.

C.2 FIXED-SUPPORT TRAINING AND RANGE INTERACTION

The paired 151,898,880-parameter decoder has 12 layers, width 768, 12 heads of dimension 64, MLP width 3072 and vocabulary 50304. Within each seed (42, 137, 256), Geo and anchored Cosh share initialization, token order, optimizer, batch and training length 256. Each arm sees 499,974,144 tokens in 7629 updates of 256 sequences. Seed 42 uses microbatch 64 with accumulation 4; the other seeds use 128×2 . Fused AdamW uses $(\beta_1, \beta_2) = (0.9, 0.95)$, weight decay 0.01, clipping 1, peak/minimum LR $6 \times 10^{-4}/6 \times 10^{-5}$ and 762 warmup steps followed by cosine decay.

FineWeb-Edu training and validation use disjoint unshuffled shards `sample/10BT/000_00000.parquet` and `sample/10BT/004_00000.parquet`, revision 87f09149ef47, with GPT-NeoX-20B tokenizer revision c292233c833e (Penedo et al., 2024; Black et al., 2022). Geo uses $b = 256$ and $\omega_k = \exp[-Rk/(K-1)]$, $R = (K-1) \log 256/K$, $K = 32$. Cosh uses anchored midpoint quantiles with $\tau = 4$ and exactly the same endpoints. Only the 30 interior coordinates differ.

Evaluation scores the final 128 target tokens at 32 fixed validation anchors. Seed 20260723 selects one end position from each of 32 equal-width stream bins with a 2048-token guard; the longest windows are disjoint. At anchor e , length H uses tokens $[e-H, e)$ and scores $[e-128, e)$. The retained-range condition keeps R fixed; the target-matched condition uses $R(H) = (K-1) \log H/K$ while retaining each trained weight set and normalized z . Both use unit gain. Training seed is the replication unit.

Table 6: Exact-range anchored EVQ-COSH minus the matched geometric control tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD
256	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008
512	-0.478	-0.271	-0.094	-0.281 $\pm$ 0.192	+0.061	+0.053	+0.067	+0.060 $\pm$ 0.007
1,024	-0.205	-0.191	-0.132	-0.176 $\pm$ 0.039	+0.182	+0.156	+0.344	+0.227 $\pm$ 0.102
2,048	-0.113	-0.181	-0.143	-0.146 $\pm$ 0.034	+0.279	+0.399	+0.701	+0.460 $\pm$ 0.218

C.3 CROSSED WEIGHTS AND RUNTIME TABLES

For seeds 137, 256, each Geo- or Cosh-trained checkpoint installs a factor-four table derived from either training allocation. The derivation uses $L_0 = 256$, $\Delta_j = 2j$ for $j = 0, \dots, 127$, and grouped causal weights $p_j = (2L_0 - 4j - 1)/(L_0(L_0 + 1)/2)$. Let X_k contain the weighted sine/cosine pair and X_{-k} the other pairs. Define

$$U_k = \frac{\|X_k - X_{-k} X_{-k}^\dagger X_k\|_F^2}{\|X_k\|_F^2}, \quad \bar{U}_k = \frac{U_k - \min_i U_i}{\max_i U_i - \min_i U_i}, \quad \omega'_k = \omega_k[1 - 3(1 - \bar{U}_k)^2/4].$$

The pseudoinverse cutoff is 10^{-10} ; zero normalization span gives $\bar{U}_k = 0$. Tables are FP32 with endpoints overwritten to ω_0 and $\omega_{K-1}/4$, and gain $1 + 0.1 \ln 4$. Evaluation uses the same anchors and tail-NLL metric at 512/1024 tokens. The 1024-token mean crossing is: Both seeds favor the

Table 7: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

Frozen weights	Geo-derived table	Anchored-EVQ-COSH-derived table
Geo-trained	3.426	5.776
Anchored-EVQ-COSH-trained	4.455	3.479

table derived from their training allocation. The seed-specific interaction contrasts are 3.400/3.251 NLL at 1024 tokens and 2.663/2.564 at 512.

C.4 MULTIPLE SHAPES AT FIXED SUPPORT

The 50.9M decoder has 6 layers, width 512, MLP width 2048 and vocabulary 50304. The 12 configurations cross bases 500K/1M, training lengths 256/1024 and 4/8/16 heads (head dimensions 128/64/32). Each of three seeds 42/137/256 trains each arm for 8,388,608 tokens, 128 updates, on a shared repeated WikiText-2 raw stream with a 1% supervised passkey mix. Execution is float32 on MPS. The preassigned arms are Geo, anchored Cosh at 0.75/1/1.25 times $d_{\text{head}}/\sqrt{L}$, and an exponential $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ whose RMS node movement matches reference Cosh without task fitting. Every arm shares sampled endpoints. The OOD mean weights ratios $r \in \{2, 4, 8\}$ by $\log_2(r + 1)$; the bootstrap unit is configuration after averaging paired seeds.

Standard-grid replication. A separate short-training study uses the 151.9M architecture, $b = 500000$, head dimension 64, length 128 and 14,974,976 tokens. Standard-grid RoPE and span-matched native-grid Cosh are evaluated on 32 test anchors separate from 16 selection anchors. Three-seed mean tail NLL for Native/Cosh is 5.5046/5.4645 at 128, 5.6563/5.5434 at 512, 5.8456/5.6966 at 1K and 6.3928/6.2935 at 8K. All three seeds favor Cosh at 512/1K/2K/4K/8K; complete recorded curves are in the bundled numerical inputs. This protocol is separate from the longer fixed-support training comparison.

C.5 FROZEN FIXED-SUPPORT AND EQUAL-DISPLACEMENT INTERVENTIONS

Frozen OLMo-2-1B and Qwen-2.5-1.5B use 64 pairs, their native fast endpoint and the native slow endpoint divided by four. Matched-support arms share gain $1 + 0.1 \ln 4$, inputs, greedy decoder and

Table 8: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times /4 \times /8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. C).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	− 0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

precision. Uniform z is log-linear between these endpoints. The derived arm computes residual fractions U_k as above over native integer separations weighted by $L - d$, grouped in pairs at rounded weighted centroids, then uses $\omega'_k = \omega_k[1 - 3(1 - \bar{U}_k)^2/4]$ and exact endpoint overwrite. The coarse arm enumerates integer linear-ramp bounds, minimizing MSE to $(1 - \bar{U}_k)^2$ with lexicographic tie-breaking: bounds are (20, 22) for OLMo and (28, 31) for Qwen. These are geometric, label-free controls.

On OLMo’s nine held-out tasks (20/task, 16K), uniform/coarse/derived score 0.56/61.04/60.47%; Qwen’s four development tasks (20/task, 64K) score 57.75/64.00/66.50%. The sampled support is identical within each three-arm block. A separate OLMo six-task follow-up uses 24/48 inputs at 4/16K, $s = 4$, band [14, 32] and gain $1 + 0.1 \ln 4$:

	MrPro	Official YaRN	MrUni	BM
4K	37.85	54.38	76.88	81.81
16K	2.78	6.94	32.12	51.32

BM and Uni have the same interior sum $(n - 1)/2$ and full-table sum 40.5; Pro’s full-table sum is $37\frac{2}{3}$. Thus BM–Uni identifies a shape contrast at equal total displacement. The newer clean T–C comparison is reported separately in Appendix D.3.

C.6 SPECTRUM-PRESERVING COORDINATE REASSIGNMENT

A block permutation obeys $\mathcal{R}_{\pi\Omega}(d) = P\mathcal{R}_{\Omega}(d)P^\top$, so jointly permuting Q/K and frequencies preserves the bilinear kernel exactly. With weights fixed, permuting only interior frequency assignments changes model performance. The historical OLMo log-law table uses $L_0 = 4096$, $b = 500000$, $K = 64$, $s = 4$, gain $1 + 0.074 \ln 4$, and fixes slots 0, 63. Its 20-document PG-19 tail NLL changes from 3.10423 to 6.86493. Qwen-2.5-1.5B recomputes the same profile at $L_0 = 32768$, $b = 10^6$; on 20 inputs per task at 64K (single-key, multikey-2, multikey-3, tracking), macro changes from 70 to 0%. These fixed-spectrum interventions isolate coordinate assignment; their table is distinct from TailSpline. Source identities remain in the numerical supplement.

D FROZEN EVALUATION: PROTOCOLS AND COMPLETE RESULTS

D.1 SHARED MODELS, TABLES, SCORING AND UNCERTAINTY

Llama-3-8B-Instruct uses native length 8192, $K = 64$, base 500000 and band [18, 35]; OLMo-2-0425-1B-Instruct uses 4096, $K = 64$, base 500000 and band [14, 32]. T/P share frozen weights, sampled endpoints, outer bands and gain $1 + 0.1 \ln 4$, while using their respective static $s = 4$ interior tables. Tables are installed in FP32 after loading BF16 weights and are fixed across layers and input lengths. Evaluation uses Flash-SDPA and greedy decoding. No table is fitted using model outputs.

Clean RULER pins revision c3f5e3b4f87f, selects source records in order, and uses neither content padding nor depth selection. Both arms use identical prompt tokens, references and generation

caps: 128 for NIAH, 30 for tracking, 120 for common words, 50 for frequent words and 32 for QA. The endpoint is the official within-task score averaged equally across the 13 tasks. All outputs, including empty and capped answers, remain in the score. Intervals use 20,000 paired resamples within task and length, with tasks fixed; source-record intervals for RULER and source-context clustered intervals for natural QA are separate estimands.

D.2 CLEAN RESULTS ACROSS LENGTH AND MODEL

Llama uses 650/650/2600 pairs at 8/16/32K; OLMo uses 2600 at 16K. All clean runs use batch 1 and 8192-token prefill chunks. Llama 32K inputs span 28270–32606 tokens; the fixed QA-1 supplement completes 200 source-order records without output-based selection. OLMo’s clean panel uses generation seed 20261101 and QA source offset 5600.

Table 9: **Complete clean task scores (%)**. N is original RoPE; T/P are TailSpline/MrPro. Llama 8/16/32K use 50/50/200 inputs per task; OLMo 16K uses 200.

Task	Llama 8K			Llama 16K		Llama 32K		OLMo 16K	
	N	T	P	T	P	T	P	T	P
cwe	97.60	95.40	89.60	83.80	80.00	1.65	0.30	7.55	3.30
fwe	87.33	9.33	4.00	25.33	34.67	61.83	39.67	51.83	23.83
MK_1	100.00	100.00	98.00	100.00	88.00	84.50	62.50	65.00	6.50
MK_2	96.00	100.00	94.00	98.00	86.00	47.00	9.00	43.00	0.50
MK_3	98.00	98.00	94.00	92.00	84.00	24.00	2.50	4.50	0.00
MQ	99.50	96.50	97.00	98.00	97.00	95.25	94.88	73.25	4.12
MV	97.50	97.00	92.50	98.50	97.50	93.12	97.62	77.12	2.25
S_1	100.00	100.00	100.00	100.00	100.00	100.00	97.00	97.50	11.50
S_2	100.00	100.00	100.00	100.00	98.00	99.50	99.00	100.00	18.00
S_3	100.00	100.00	100.00	100.00	100.00	100.00	94.00	64.00	0.00
qa_1	70.00	70.00	72.00	76.00	70.00	64.00	55.50	39.50	31.50
qa_2	32.00	46.00	44.00	60.00	50.00	37.00	27.50	25.50	18.50
vt	92.40	94.80	86.00	87.60	90.00	79.60	55.60	9.70	0.00

Llama 32K T-P is +11.72 points, interval [10.32, 13.11]. Family differences (retrieval/tracking/aggregation/QA) are +10.86/ + 24.00/ + 11.76/ + 9.00 points, with intervals [9.11, 12.61]/[20.60, 27.40]/[8.80, 14.77]/[4.25, 13.75]. Leaving out any one task keeps the macro difference between 9.53 and 13.07 points. At 16K, the Full-13 difference is +3.39 [1.53, 5.34]; family differences are +4.50/ − 2.40/ − 2.77/ + 8.00. The latter QA interval is [1, 15] and retrieval interval [2.50, 6.69].

OLMo clean 16K T-P is +41.42 points, interval [39.99, 42.81]. Its family differences are +60.19/ + 9.70/ +16.13/ +7.50 points, with intervals [58.30, 62.03]/[7.00, 12.60]/[13.86, 18.42]/[2.75, 12.25]. All thirteen task means increase. T/P end with EOS on 1251/1392 inputs and hit the cap on 1349/1208, with no empty outputs.

At native 8K, original RoPE uses the native table and unit gain. T/P/original RoPE score 85.1564/82.3923/90.0256%. T-P is +2.7641 [1.2923, 4.2538] points; T-original RoPE is −4.8692 [−6.5718, −3.1462], a 5.41% relative reduction. The difference concentrates in FWE: T/P have 45/48 empty responses, all on that task, while original RoPE has none. Across all 650 inputs, T/P/original RoPE EOS counts are 648/644/645 and cap counts 2/6/5. The other twelve tasks have a descriptive T-original RoPE mean of +1.225 points; the primary Full-13 endpoint retains FWE.

D.3 CLEAN EQUAL-DISPLACEMENT COMPARISON

C is the analytic profile in Appendix B.2. T/C share outer bands, endpoints, gain and total log-frequency displacement; the deployed FP32 $\sum m_k$ discrepancy is below 10^{-8} . Completed clean T/P generations are reused; C adds the same 650/2600 inputs at 16/32K with batch 1 and 8192-token prefill chunks. T/C score 86.0949/86.3462% at 16K, difference −0.2513 [−1.4051, 0.8949] points; at 32K they score 68.2660/66.1699%, difference +2.0962 [1.1134, 3.0769] points. At 32K, T and C both have EOS/cap rates 98.96/1.04%. Their full-pair ranks are 10.0773/10.0926, compared with P’s 10.2149; the task contrast therefore also distinguishes positional rank from learned use.

Table 10: **Complete clean shape comparison.** Scores (%) at 16K (50/task) and 32K (200/task).

Task	T16	C16	P16	T32	C32	P32
cwe	83.80	83.20	80.00	1.65	0.65	0.30
fwe	25.33	30.00	34.67	61.83	54.83	39.67
multikey_1	100.00	98.00	88.00	84.50	88.00	62.50
multikey_2	98.00	98.00	86.00	47.00	36.50	9.00
multikey_3	92.00	92.00	84.00	24.00	9.50	2.50
multiquery	98.00	98.50	97.00	95.25	95.25	94.88
multivalue	98.50	98.00	97.50	93.12	94.88	97.62
single_1	100.00	100.00	100.00	100.00	99.50	97.00
single_2	100.00	100.00	98.00	99.50	98.50	99.00
single_3	100.00	100.00	100.00	100.00	100.00	94.00
qa_1	76.00	76.00	70.00	64.00	69.50	55.50
qa_2	60.00	60.00	50.00	37.00	37.00	27.50
vt	87.60	88.80	90.00	79.60	76.10	55.60

D.4 NATURAL QA AND LONGBENCH V2

The frozen Natural-QA pool has 631 questions from 524 source-context clusters across HotpotQA, 2WikiMQA, Qasper, NarrativeQA and MultiFieldQA-en (Bai et al., 2024). Membership was defined using the OLMo > 4096-token threshold and retained under Llama retokenization. Actual inputs are 4110–16319 for OLMo and 3694–16304 for Llama. Llama has 316 questions within its native window and 315 above it. Generation caps are 32 for HotpotQA/2Wiki, 128 for Qasper/NarrativeQA and 64 for MultiFieldQA. Batch is 1; Llama uses 8192-token chunked prefill and OLMo direct prefill.

The primary score averages whole-response normalized token F1 within each task, then equally across tasks. The paired bootstrap resamples source-context clusters within task, dividing sampled score sums by sampled question counts; 20,000 draws use seed 20260914 for Llama and 20260930 for OLMo. There are no shared clusters across tasks. Both arms retain every answer.

Table 11: Natural-QA631 task means and complete-output health.

Task	N	TailSpline (%)	MrPro (%)	Difference (pp)
hotpotqa	166	52.98	53.09	-0.10
2wikimqa	173	36.96	35.19	+1.76
qasper	119	40.73	41.04	-0.31
narrativeqa	61	22.82	23.77	-0.95
multifieldqa_en	112	51.91	51.33	+0.58
EOS	—	625	624	—
Cap-hit	—	6	7	—
Empty	—	0	0	—
Generated tokens	—	6972	7019	—

Table 12: **OLMo Natural-QA task breakdown.** Whole-response F1 (%); each task has equal weight in the primary mean.

Task	Questions	TailSpline	MrPro	Δ (pp)
hotpotqa	166	34.65	28.63	+6.02
2wikimqa	173	26.29	21.55	+4.74
qasper	119	18.03	13.94	+4.09
narrativeqa	61	9.85	9.51	+0.34
multifieldqa_en	112	35.80	34.49	+1.31

Llama’s overall difference is +0.1957 [−1.5311, 1.8864] points; its within/above-native contrasts are +0.6998 [−1.2478, 2.8073] and −0.9383 [−4.1494, 1.9539]. Document-equal sensitivity gives

T/P 40.8141/40.8375%. OLMo’s overall difference is +3.2994 [0.3359, 6.4028] points, with every task mean positive; question-equal sensitivity gives +3.9194 [1.1977, 6.6794]. Its 4K–8K and 8K–16K strata contain 316/315 questions and give +6.3989 [1.8747, 10.8706] and +1.5095 [−4.9432, 7.6943] points. OLMo T/P EOS/cap/empty counts are 567/581, 64/50, 0/0.

LongBench v2 uses 117 complete Llama inputs between 8K and 32K, with its direct-answer multiple-choice adapter, batch 1, matched caps, static $s = 4$ tables and 8192-token prefill chunks. Correct parsed options score one; incorrect/unparseable answers score zero. T/P answer 41/36 correctly, giving 35.0427/30.7692% and difference +4.2735 [−3.3906, 11.9658] points under source-context cluster resampling. T alone succeeds on 13 inputs, P on 8, with 96 ties. The 28 inputs at 8K–16K

Table 13: **LongBench v2 complete-input subset**. Accuracy (%) with actual Llama-tokenized input lengths between 8K and 32K.

Domain	Inputs	TailSpline	MrPro
Code repository	3	33.33	0.00
In-context learning	7	14.29	14.29
Structured data	1	0.00	0.00
Dialogue history	12	41.67	16.67
Multi-document QA	33	39.39	30.30
Single-document QA	61	34.43	37.70

give T/P 39.29/35.71%; the 89 above 16K give 33.71/29.21%. Overall EOS/cap/empty counts are 88/91, 29/26, 0/0. Accuracy and token F1 remain separate metrics.

D.5 CLASSIC LENGTH CURVES AND SUPPLEMENTARY CONTROLS

Classic Llama and OLMo use 10 RULER inputs/task at $L, 2L, 4L$ (390/arm), with output-independent depth selection and prefix padding. Single-evidence prompts have two examples at each 10/30/50/70/90% depth; multi-evidence prompts alternate dispersed and middle-cluster targets. A repeated irrelevant-record prefix fills the input to cap minus 128 tokens. The shared task-specific decoding budgets above are retained. Preparation seeds are 20260924/20260925 (Llama/OLMo), with fixed length/task offsets; the bundled entrypoints specify the deterministic construction.

PPL uses 32 Proof-Pile arXiv test documents and 14 PG-19 test-book prefixes (Azerbayev et al., 2022; Rae et al., 2019). At each length it is the exponential of summed whole-prefix loss divided by scored tokens. Normalized trapezoidal log-length AUC has weights (1/4, 1/2, 1/4); PPL AUC integrates PPL itself. The document bootstrap resamples document IDs jointly across lengths; task bootstrap resamples within task-length cells.

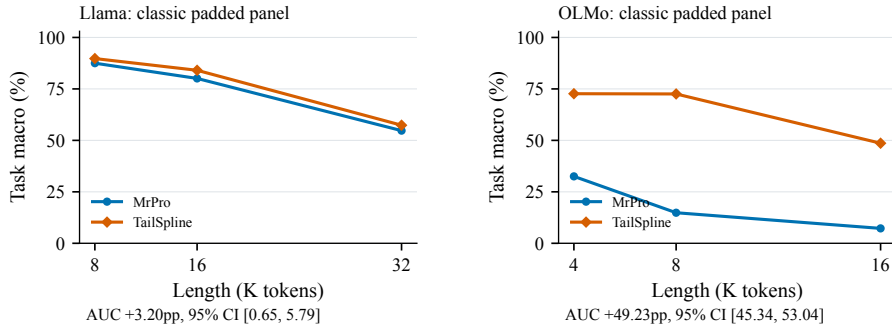


Figure 6: **Classic task and PPL curves**. Each model retains its own recorded depth-selected, prefix-padded protocol.

Llama Full-13 AUC T/P is 78.8013/75.6026%, difference +3.1987 [0.6538, 5.7853] points; OLMo is 66.60/17.36%, difference +49.23 [45.34, 53.04]. Llama PPL at 8/16/32K is T 5.295283/4.530318/4.110622, P 5.286429/4.518111/4.161831. Its NIAH AUC improves by 3.75 points; QA AUC changes by −1.25. Llama EOS/cap/empty counts are T/P 388/384, 2/6, 30/37.

The original classic T-C diagnostic gives AUC 78.8013/79.2115%, difference -0.4103 $[-2.6282, 1.8205]$ points, with batch 1/2. A 39-input batch check has identical scores but six different generated sequences. This recorded diagnostic is separate from the clean comparison. The classic 130-input native reference scores 91.8846%, versus T 89.7436%, difference -2.1410 $[-6.1410, 1.9231]$ points. Whole-prefix 8K PPL on 46 documents is original RoPE/T/P 5.275919/5.295283/5.286429; T-original RoPE is $+0.019364$ with paired document interval $[0.006602, 0.034861]$.

E NATIVE-WINDOW ALLOCATION: CONSTRUCTION AND COMPLETE RESULTS

E.1 REFERENCE OBJECTIVE AND NUMERICAL CONSTRUCTION

NCP improves task performance within the native context by reallocating the interior frequencies while fixing both sampled endpoints. Let $\phi_k = (L-1)\omega_k$ and $\omega'_k = \omega_k e^{-u_k}$. The reference problem is a binary target-distractor comparison with unit amplitudes, aligned target phase and uniformly random distractor phase:

$$r(\theta) = \frac{1}{2\pi} \int_0^{2\pi} \log(1 + \exp(\cos \psi - \cos \theta)) d\psi, \quad (39)$$

$$\mathcal{R}(\phi) = 2 \int_0^1 (1-t)r(\phi t) dt. \quad (40)$$

The triangular distance weight is the distribution of absolute separation between two uniform positions on a continuous unit interval. The aligned reference comparison supplies a public design objective. The Transformer task scores below evaluate the resulting fixed construction.

We minimize the objective in Eq. (10), with fixed $\lambda = 9/4$ (quadratic coefficient $\lambda/2 = 9/8$), subject to

$$u_0 = u_{K-1} = 0, \quad 0 \leq u_k \leq 2/9, \quad \log \frac{\omega'_k}{\omega'_{k+1}} \geq \frac{1}{2} \log \frac{\omega_k}{\omega_{k+1}}. \quad (41)$$

The endpoint constraints keep support fixed; the gap condition preserves ordering and at least half each original adjacent log gap. Writing $r(\theta) = a_0 + \sum_{j \geq 1} a_j \cos(j\theta)$ gives

$$\mathcal{R}(\phi) = a_0 + \sum_{j \geq 1} a_j \operatorname{sinc}^2(j\phi/2), \quad \operatorname{sinc}(x) = \frac{\sin x}{x}, \quad \operatorname{sinc}(0) = 1. \quad (42)$$

This follows by integrating each cosine against $2(1-t)$. With $q(\phi) = d\mathcal{R}/d \log \phi$, an interior scalar solution satisfies $\lambda u = q(\phi e^{-u})$. We compute r on a 1024-point angular grid, retain 32 Fourier modes and solve the scalar stationary equations by bisection, using zero shift when the native gradient is nonpositive. The two endpoint shifts are reset to zero. On the deployed OLMo grid all resulting log-gap constraints are inactive, so the scalar solutions are feasible jointly.

The deployed grid has 64 pairs, base 500000 and native length 4096. It changes 46 FP32 entries, reaches maximum relative slowdown 16.7164% and retains a minimum log-gap ratio of 0.7681. The maximum scalar stationarity residual is below 10^{-11} . The mean reference risk decreases from 0.614082 to 0.609641; the penalized objective is 0.611853. These are reference-model quantities, separate from model task scores. The table and unit gain are fixed across layers and inputs; construction reads no checkpoint weights, activations or evaluation outputs. The source archive includes the original constructor and an exact FP32 table reproduction check.

E.2 NATIVE-CONTEXT TASK RESULTS AND COMPANION CONTROLS

The OLMo native 4K development panel contains 60 source-order inputs per RULER task, 780 total, without content padding. It is the same frozen panel used for the companion allocation studies. NCP and original RoPE share the checkpoint, unit gain, inputs, BF16 precision, greedy decoder, batch 1 and 4096-token prefill chunks. Their scores are 70.7821/69.3739%; the paired task-equal difference is $+1.4081$ points, interval $[0.2009, 2.5833]$. NCP has higher per-input scores on 62 inputs, original RoPE on 31, with 687 ties. Every task remains in the aggregate. Companion half-turn

Table 14: **NCP within the native 4K window.** Clean Full-13, 60 inputs per task.

Task	Original RoPE	NCP	Δ (pp)
cwe	7.83	4.67	-3.17
fwe	57.78	60.00	+2.22
multikey_1	81.67	78.33	-3.33
multikey_2	91.67	95.00	+3.33
multikey_3	58.33	61.67	+3.33
multiquery	89.17	90.83	+1.67
multivalue	95.42	98.33	+2.92
single_1	100.00	100.00	+0.00
single_2	100.00	100.00	+0.00
single_3	98.33	98.33	+0.00
qa_1	68.33	61.67	-6.67
qa_2	35.00	30.00	-5.00
vt	18.33	41.33	+23.00

contraction and its reverse score 69.5085/67.8974%. Their exact mean contrast is +1.6111 points with paired interval [0.2713, 2.9722]; contraction relative to original RoPE is +0.1346 points with interval [-0.6603, 0.9445]. A separate historical checkpoint-calibrated V1 table scores 72.3846%, a +3.0107-point contrast with interval [1.2949, 4.7052]. V1’s calibration distinguishes it from NCP’s public-parameter construction. Point differences here are computed from arm means; the companion report’s bootstrap intervals are retained.

F LEARNING AND EXTRAPOLATION WITH COSH

F.1 MULTI-HEAD LATENT ATTENTION

We report allocation at one fixed rotary budget in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024a), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified $\tau = 1.414$ setting. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree.

Evaluation corpus and aggregation. The evaluator reads the shared cached FineWeb-Edu stream `val_fineweb-edu_5000000.pt`, nominally five million GPT-2 tokens. For each method and seed it resets NumPy RandomState to 9999, iterates through lengths 8192, 16384, 20480, 24576, 28672, and 32768, and draws eight distinct start offsets per length. Every chunk scores all $L - 1$ next-token targets, with full-sequence cross-entropy and no tail restriction. Its per-model PPL is the exponential of the mean chunk NLL; the table then takes the arithmetic mean and sample standard deviation of the three per-seed PPLs. Thus the requested targets per model are $8(L - 1)$ at each length; starts are sampled without replacement, but overlapping windows are possible. The same offsets are reused across methods.

All methods use the same fixed FineWeb-Edu evaluation cache and paired chunk offsets. The archived result JSON retains every seed and length, allowing the reported means, seed variation and scaling comparisons to be recomputed.

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

EVQ-COSH reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by +0.9%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond, showing that the allocation also composes with this wavelength-dependent extension.

Table 15: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

F.2 SHARED-START CONTINUATION AND MATCHED ADAPTATION

The 750M pair continues a common 2K geometric checkpoint for 500M tokens at 4K, with 3% passkey and 10% downstream mix (seed 42). Cosh uses $\tau = 1.5$, $K = 32$, inclusive nodes $u_k = k/(K - 1)$ and $\omega_k = \exp[-(K - 1) \log(500000)Q_{1.5}(u_k)/K]$, retaining Geo endpoints. Both arms update all weights. Greedy answer-token exact match generates the expected passkey tokens and closing marker; it does not require a following EOS.

Table 16: 750M continued-pretraining check (seed 42; 2K $\rightarrow$ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials per length. The pooled AR row combines 2K, 4K and 8K (120 trials: Geo 80/120, EVQ-COSH 111/120).

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Answer-token exact @8K	0%	77.5%
Pooled answer-token exact (2K/4K/8K)	66.67%	92.5%

For MLA and 750M continuation, AdamW uses (0.9, 0.95), $\epsilon = 10^{-8}$, weight decay 0.1, clipping 1, no dropout, peak LR 2×10^{-4} , 2% linear warmup and cosine decay to 0.1 of peak. Batch sizes are 6 (MLA) and 14 (continuation: microbatch 7, accumulation 2). Text packing is contiguous without cross-document masking. MLA uses unanchored midpoint Cosh; the continuation fixes sampled endpoints.

Matched OLMo routing continuation. OLMo-2-0425-1B-Instruct uses fixed standard-grid Cosh with $\tau = 2$, $b = 500000$ and rank-64, alpha-128 Q/K/V/O LoRA. Native/Cosh share checkpoint, training rows, budgets and evaluation inputs. Starting from each arm’s 300-step routing parent, a matched 100-step query-gap continuation uses peak LR 5×10^{-5} and 20 warmup steps; a 32-step answer-plus-EOS stage uses 10^{-5} and 4 warmup steps. Both use seed 20260728, AdamW (0.9, 0.95), zero weight decay, clipping 1, BF16, batch 8 (micro 4, accumulation 2), and cosine LR decay.

Each cycle has two counterfactual-routing updates and one natural-text update. The 1024 training pairs and 128 calibration pairs use different seeds and query templates. Examples contain eight key-value blocks; changing the queried value changes the target answer. Loss combines answer CE and a counterfactual margin, adding immediate EOS CE of weight one in the final stage. Physical training inputs stay at most 4096 tokens; explicit position IDs expose target-range phases. Query offsets follow $16 \lfloor (211i \bmod 12300)/16 \rfloor$; final source-to-answer gaps are $4608 + (211i \bmod 10753)$. The bundled schedule verifier reproduces these policies and the complete source receipts.

Evaluation uses real contiguous numeric-NIAH prompts, 100 per length, and requires the complete expected answer followed by terminal EOS. Native/Cosh scores are 95/100 versus 100/100 at 4K, 18/100 versus 98/100 at 8K, and 0/100 versus 60/100 at 16K. All outputs terminate with EOS. The Cosh long-gap subsets score 48/50 at 8K and 31/66 at 16K. This is matched learned adaptation, with explicit long-phase exposure, distinct from the frozen allocations.